PSGR
Krishnammal College for Women

DESIGNING AND DEPLOYING COMPUTER NETWORKS

FINAL PROJECT SUBMITTED IN PARTIAL FULFILLMENT OF THE

AWARD OF THE DEGREE

OF

BACHELOR OF INFORMATION TECHNOLOGY

SUBMITTED BY

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Coimbatore — 641004

FEBRUARY 2024

CERTIFICATE

This is to certify that this project work entitled "**DESIGNING AND DEPLOYING COMPUTER NETWORKS**" submitted to PSGR Krishnammal College for Women,Coimbatore, in partial fulfilment of the requirement for the award of the Bachelor of Information Technology, is a record of the original work done by **NITHYASHREE R S (22BIT071**) during her period of the study in Department of Information Technology, PSGR Krishnammal College for Women, Coimbatore under my supervision and guidance and her project work has not formed the basis for the award of any Degree/Diploma/Associate/Fellowship or similar title to any candidate of any University.

Forwarded by

Dr. G SOPHIA REENA M.C.A., M.Phi1., Ph.D.,

Head of the Department

DECLARATION

I hereby declare that the final project entitled **"DESIGNING AND DEPLOYING COMPUTER NETWORKS"** submitted to PSGR Krishnammal College for Women, Coimbatore for the award of the Degree of Bachelor of Information Technology, is a record of original work done by **NITHYASHREE R S (22BIT071)** under the supervision and guidance of **Dr. G SOPHIA REENA M.C.A., M.Phil., Ph.D., Associate Professor & Head, Department of Information Technology,** PSGR Krishnammal College for Women, Coimbatore and that this final project work has not formed the basis for the award of any Degree/Diploma/Associate Fellowship or similar title to any candidate of any University.

NITHYASHREE R S
(22BIT071)

Endorsed by

Place : Coimbatore

Faculty Guide

Date :

Dr. G SOPHIA REENA M.C.A., M.Phi1., Ph.D.,

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ABSTRACT

The "Designing and Deploying Computer Networks (DDCN)" training program is designed to provide a comprehensive blend of theoretical understanding and hands-on practice in modern networking. It equips participants with the skills to design, configure, and manage robust network infrastructures, using industry-standard tools like Cisco Packet Tracer and MikroTik WinBox.

This training covers essential networking concepts, including IP addressing, VLANs, static and dynamic routing, DHCP, NAT, and wireless configurations. Participants gain practical experience in building, testing, and troubleshooting networks, beginning with foundational tasks such as cabling and tool usage and progressing to advanced configurations like network segmentation and secure wireless setup.

A distinctive highlight is the final project, which challenges participants to design and deploy a functional network integrating the principles learned. This hands-on task ensures a deep understanding of real-world scenarios, fostering critical problem-solving and application skills.

The program also emphasizes teamwork and documentation, preparing participants for collaborative professional environments. It caters to a wide audience, from beginners seeking a strong foundation in networking to experienced professionals looking to refine their expertise and stay updated with the latest tools and technologies.

By completing the DDCN program, participants gain the confidence and capability to address modern networking challenges, positioning themselves as skilled professionals ready to contribute to designing and managing advanced network infrastructures.

1. INTRODUCTION

Networking is the backbone of modern communication, enabling seamless data exchange across devices and systems. The "Designing and Deploying Computer Networks (DDCN)" training program equips participants with the skills to design, configure, and maintain advanced networks. Covering fundamentals like cabling and IP addressing to advanced topics such as VLANs, routing, and NAT, the program emphasizes practical learning with tools like Cisco Packet Tracer and MikroTik WinBox. A final project integrates learned concepts into a functional network solution, fostering real-world problem-solving skills. By the end of this training period, participants will be well-equipped to tackle real-world networking challenges, making this training ideal for those aspiring to excel in the fields of network administration and design.

1.1MODULES AND DESCRIPTION:

- VLANs and Trunking
- Data VLAN
- Voice VLAN
- Native VLAN
- Management VLAN
- Trunk links
- Inter-VLAN Routing
- Testing Inter-VLAN
- Troubleshooting VLANs and Inter-VLAN Routing

VLANs and Trunking

- A **VLAN** (Virtual Local Area Network) allows you to split one physical network into several smaller, isolated networks, improving security and making it easier to manage network traffic. Think of it like dividing an office into different departments, where each department can only communicate with its members unless allowed otherwise.
- **Trunking** is the method used to send data from multiple VLANs through a single connection, usually between switches or between a switch and a router. It helps reduce the number of physical cables needed by allowing all VLANs to share the same cable while keeping the traffic for each VLAN separate.

Data VLAN and Voice VLAN

- A **Data VLAN** is used to carry regular network traffic, like emails, web browsing, and file sharing. It's what most computers, printers, and other office devices connect to. By placing this traffic in its own VLAN, it's easier to manage and keep separate from other types of network traffic.
- A **Voice VLAN** , on the other hand, is designed specifically for VoIP (Voice over IP) phones, which handle phone calls over the internet. Separating voice traffic from data traffic ensures clearer, uninterrupted calls, as the network can give priority to voice packets. This way, phone calls remain smooth even if the data network is busy

Management VLAN

- A **Management VLAN** is a special VLAN used for managing network devices like switches and routers. It separates the network management traffic (like configuring devices or monitoring the network) from regular user traffic, adding an extra layer of security. For example, network admins can use the management VLAN to securely access and control devices without mixing with other network traffic.

Native VLAN

- The **Native VLAN** is the VLAN that doesn't have any tags or labels when traffic moves across a trunk link (a connection carrying multiple VLANs). It's often used for backward compatibility with older devices that don't understand VLAN tagging. By default, untagged traffic is assigned to the native VLAN, but it can be changed as needed.

Trunk links

- A **trunk link** lets data from different VLANs travel over a single cable between switches or between a switch and a router. It's like a highway with different lanes for each VLAN, and they stay separate. Trunk links use tags (usually IEEE 802.1Q) to identify which VLAN the data belongs to, making network management simpler and reducing the need for multiple cables.

Inter-VLAN Routing

- **Inter-VLAN routing** lets devices in different VLANs communicate. Normally, devices in the same VLAN can talk to each other, but not to those in other VLANs. A router or Layer 3 switch is used to forward traffic between VLANs, allowing communication, like a translator helping people speak different languages. This is helpful when different parts of the network, like departments, need to interact.

Testing Inter-VLAN

- **Testing Inter-VLAN** routing involves checking if devices in different VLANs can communicate. First, confirm that the router or Layer 3 switch connecting the VLANs is properly configured. Then, use tools like ping or traceroute to test connections by trying to ping a device's IP address in another VLAN. If the ping is successful, Inter-VLAN routing is working; if it fails, you may need to troubleshoot the settings or check for network configuration issues.

Troubleshooting VLANs and Inter-VLAN Routing

- **Troubleshooting VLANs and Inter-VLAN routing** means fixing problems related to network segmentation and communication between different VLANs. Start by checking if VLANs are set up correctly on switches and ensure the right ports are assigned. For Inter-VLAN routing, make sure the router or Layer 3 switch is configured properly and that the interfaces for each VLAN are active. Common troubleshooting steps include using commands to check VLAN status, pinging devices for connectivity, and ensuring trunk links can carry traffic for all VLANs. By checking these areas, you can solve communication issues in the network

1.2 ORGANIZATION PROFILE

INTERNET SOCIETY CHENNAI CHAPTER is a leading firm specializing in cyber security solutions for the industry/sector. Established in 1992, the company is known for an independent, international, non-profit, cost-based organization established in 1992. We are dedicated to the stability, continuity and advancement of the Internet, not for its own sake but rather for the benefits the Internet can bring to all people. As an At Large Structure, ISOC India Chennai ALS pursues Internet User's interests by taking part and contributing to policy discussions at the national and global level, including participation in ICANN meetings, Internet Governance Forums. We are dedicated to the stability, continuity and advancement of the Internet, not for its own sake but rather for the benefits the Internet can bring to all people. The ISOC Chennai At-Large Structure was formed as a division of the ISOC Chapter with volunteers from among the Members of the ISOC Chapter. The Internet Society supports and promotes the development of the Internet as a global technical infrastructure, a resource to enrich people’s lives, and a force for good in society. Our work aligns with our goals for the Internet to be open, globally connected, secure, and trustworthy. We seek collaboration with all who share these goals. Together, we focus on: Building and supporting the communities that make the Internet work Advancing the development and application of Internet infrastructure, technologies, and open standards, and advocating for policy that is consistent with our view of the Internet.

2. SYSTEM STUDY

2.1 EXISTING SYSTEM

The existing system for IP address allocation in network environments relies heavily on manual configuration. In this setup, administrators must assign IP addresses, subnet masks, default gateways, and DNS server addresses individually to each device. This approach is time-consuming, especially in networks with a large number of devices, as each configuration must be inputted manually. Static IP addressing is the norm, which means that every device retains the same IP address unless manually reconfigured. While this method offers some level of control, it introduces significant challenges. Errors, such as assigning duplicate IP addresses or incorrect subnet masks, are common and can disrupt network functionality. Moreover, in networks where devices frequently connect and disconnect, such as in dynamic office environments or large-scale organizations, the manual allocation system struggles to adapt to the constant changes, leading to inefficiencies and downtime.

- **Scalability:** As the number of devices grows, maintaining a static IP system becomes increasingly impractical. Managing multiple subnets, for example, requires advanced planning and careful coordination to avoid conflicts. Furthermore, without automation, expanding the network or introducing new devices requires manual intervention, slowing down the process.
- **Lack of flexibility:** It is particularly problematic in scenarios where time-sensitive changes are necessary, such as during office expansions or when integrating new hardware.

2.2 PROPOSED SYSTEM

The proposed system introduces the use of a DHCP (Dynamic Host Configuration Protocol) server configured on a Cisco router to automate the allocation of IP addresses and related settings. Unlike the existing manual approach, DHCP dynamically assigns IP addresses to devices within a defined range, streamlining the process and eliminating the need for individual configuration. This system leverages a centralized DHCP server, which simplifies IP management by automatically tracking assigned addresses and releasing them when devices disconnect from the network. By doing so, the system prevents IP address conflicts and optimizes resource utilization. The implementation of DHCP saves time and significantly reduces the administrative overhead associated with manual configuration.

Additionally, the proposed system is highly scalable and adaptable, making it suitable for both single-subnet and multi-subnet networks. In multi-subnet environments, the router's sub-interfaces are configured to support VLANs, with each VLAN assigned its own DHCP pool. This ensures that devices in different subnets receive appropriate IP addresses, enabling seamless communication and maintaining network efficiency. The system also supports flexibility for future growth, as new devices or subnets can be easily integrated by updating the DHCP configuration. Automation minimizes human errors, such as duplicate IP allocations or incorrect settings, thereby enhancing network stability and reliability.

- **Centralized DHCP Management and IP Address Reservation:** A single DHCP server allocates IP addresses for all devices in the network. Specific IPs reserved for critical devices (e.g., servers, printers).
- **Subnet Support Scalability:** IP allocation across multiple subnets using DHCP relay. Ability to add more devices or subnets without reconfiguring the DHCP server extensively.

3. SYSTEM SPECIFICATION

2.1 HARDWARE SPECIFICATION:

Processor	:	Intel P-IV system
Processor Speed	:	250MHzto 833MHz
RAM	:	8GB RAM
Display	:	1366 x 768 resolution or higher

2.2 SOFTWARE SPECIFICATION:

Operating System	:	Windows 10
Packet Tracer Version	:	Cisco Packet Tracer 8.x
Disk Space	:	3 GB free storage for installation

2.3 NETWORKING COMPONENTS AND DESCRIPTION

- Cable Stripper
- Crimping Tool
- LAN (Local Area Network) Tester
- RJ-45 Jack (Ethernet Jack)
- Punch-Down Tool
- Network Cable

Cable Stripper

- **Function:** A cable stripper is used to remove the outer insulation from cables, typically Ethernet cables like Cat5e, Cat6, or Cat7.
- **Usage:** It allows you to expose the inner wires without damaging them, making it easier to prepare the cable for termination or connection to a jack.

Crimping Tool

- **Function**: A crimping tool is used to attach connectors, like RJ-45, to the ends of network cables.
- **Usage**: After stripping the cable and arranging the wires according to the required pinout (e.g., T568A or T568B standard), the crimping tool is used to press the RJ-45 connector onto the cable, securing the wires in place.

LAN (Local Area Network) Tester

- **Function:** A LAN tester is used to verify that a network cable is wired correctly and that it can transmit data effectively.
- **Usage:** After crimping the connectors onto both ends of the cable, you can use the LAN tester to check for continuity, shorts, miswiring, and other issues. It usually consists of two parts: a transmitter and a receiver. You connect one end of the cable to the transmitter and the other end to the receiver to test the connection.

RJ-45 Jack (Ethernet Jack)

- **Function:** RJ-45 jacks are the standard connectors used for Ethernet networking. They connect twisted-pair network cables to network devices like computers, routers, and switches.
- **Usage:** These jacks can be found in wall outlets or patch panels and are used to establish wired network connections. RJ-45 connectors are terminated using a crimping tool and require precise arrangement of the wires to ensure a proper network connection.

Punch-Down Tool

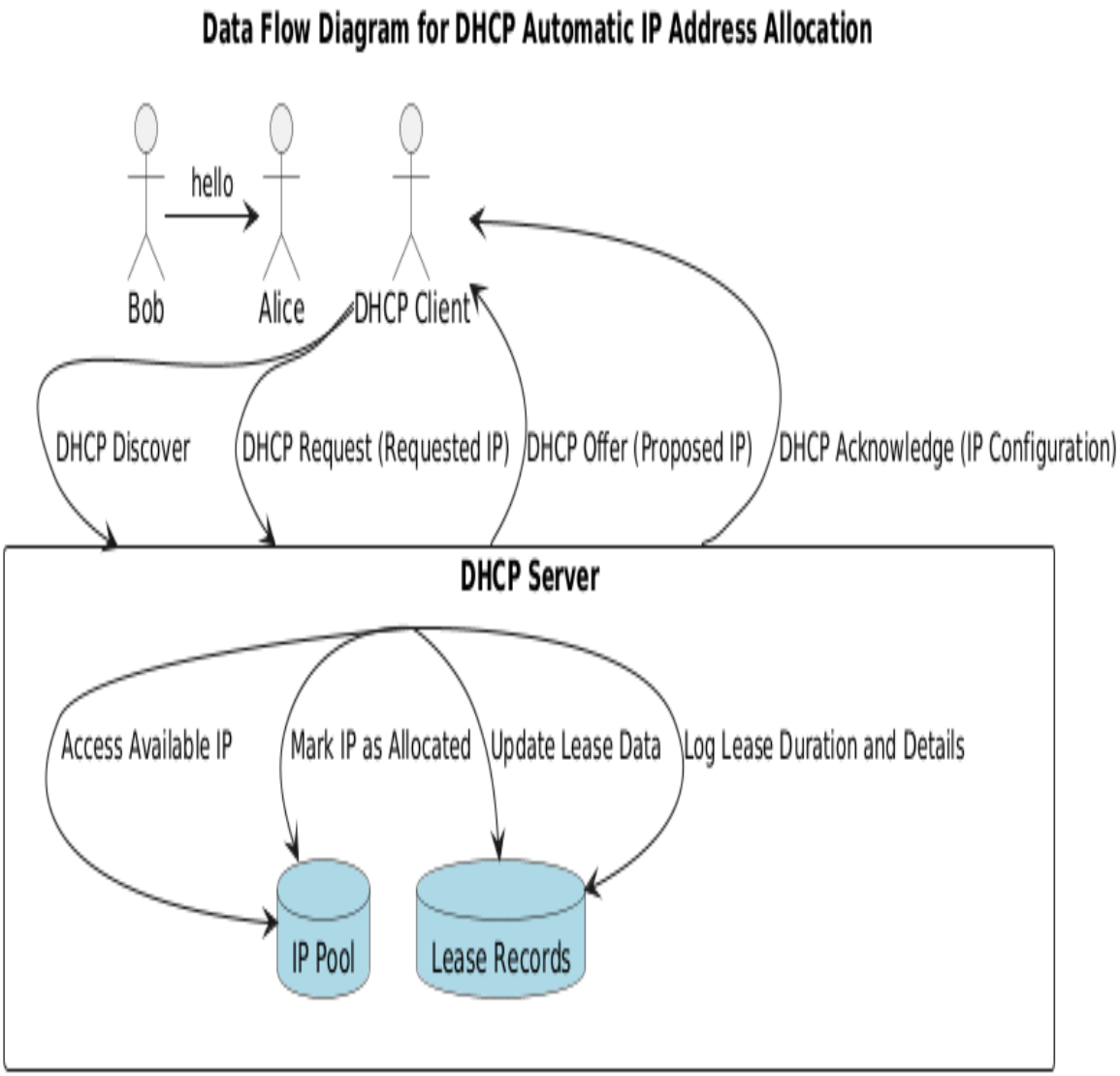
- **Function and Usage:** A punch-down tool is used to terminate wires into punch-down blocks, keystone jacks, or patch panels. The tool is designed to push the individual wires into the insulation displacement connectors (IDC) of a jack or panel.

Network Cable

- **Function:** These are the actual cables used to connect devices in a network, typically Cat5e, Cat6, or Cat7 cables.
- **Usage:** Network cables are used to connect devices such as computers, routers, switches, and other network equipment. The cable consists of twisted pairs of wires that transmit data across the network.

4. SYSTEM DESIGN

4.1 DATAFLOW DIAGRAM



4.2 SYSTEM FLOW DIAGRAM

4.5 INPUT DESIGN

i. SINGLE SUBNET NETWORKING:

COMPONENTS	COMPONENT TYPES	NO.OF. COMPONENTS
Router	Router [1941]	1
Switch	Switch [2960-24TT]	1
PC	PC [PC-PT]	5

ROUTER CONFIGURATION

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.9
Router(config)#ip dhcp pool LAN_POOL
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#exit
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

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ii. MULTIPLE SUBNET NETWORKING:

COMPONENTS	COMPONENT TYPES	NO.OF. COMPONENTS
Router	Router [1941]	1
Switch	Switch [2960-24TT]	2
PC	PC [PC-PT]	6

SWITCH COUNT	PC COUNT	CONNECTING VLAN's
Switch 0	PC [0 – 2]	VLAN 10
Switch 1	PC [3 – 5]	VLAN20

ROUTER CONFIGURATION

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp pool VLAN_10_POOL
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool VLAN_20_POOL
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#interface g0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
```

```
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#exit
```

SWITCH 1 CONFIGURATION

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name SALES
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name MARKETING
Switch(config-vlan)#exit
Switch(config)#interface range f0/3 - 5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface f0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch(config-if)#
```

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Top

SWITCH 2 CONFIGURATION

```
Switch>enable
Switch#conf
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Configuring from terminal, memory, or network [terminal]? terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name SALES
Switch(config-vlan)#Exit
Switch(config)#vlan 20
Switch(config-vlan)#name MARKETING
Switch(config-vlan)#exit
Switch(config)#interface range f0/3 - 5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface f0/2
Switch(config-if)#switchport mode trunk
Switch(config-if)#
Switch(config-if)#
```

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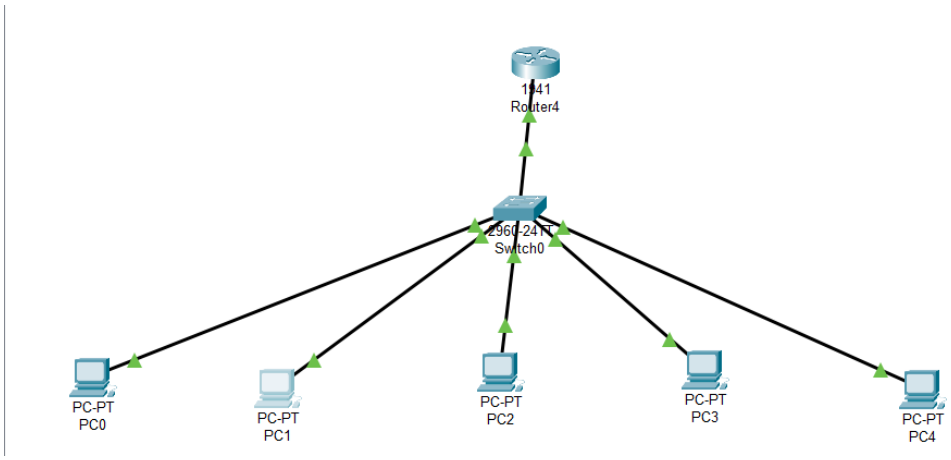
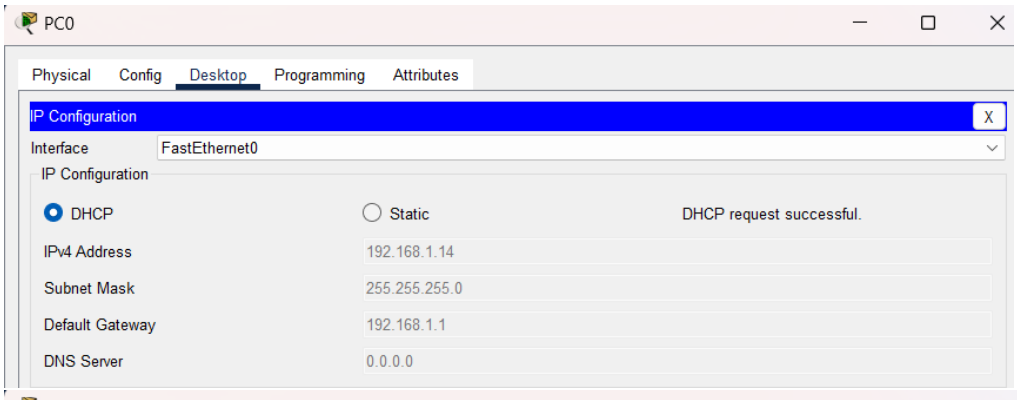
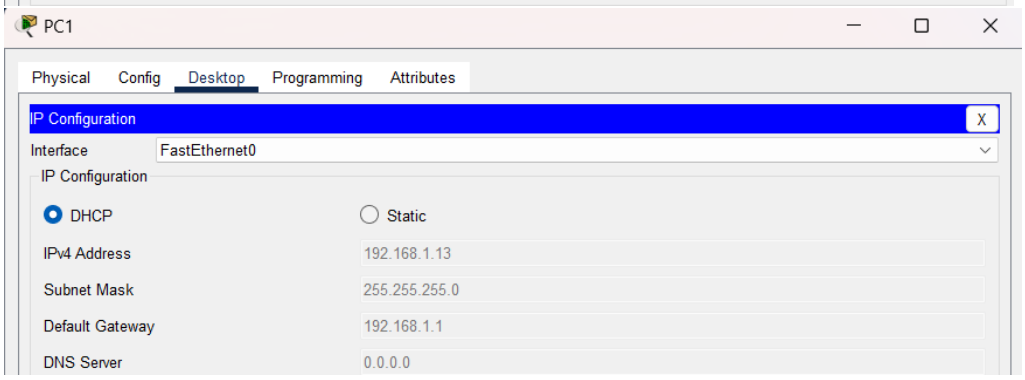
4.6 OUTPUT DESIGN

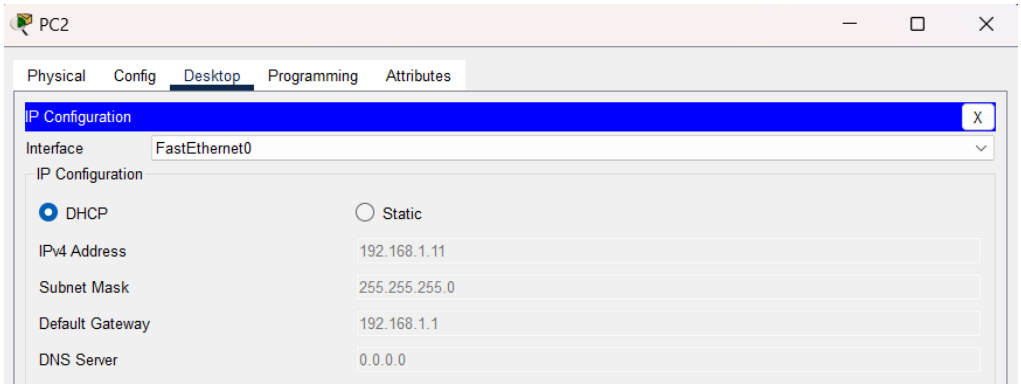
i. SINGLE SUBNET NETWORKING:

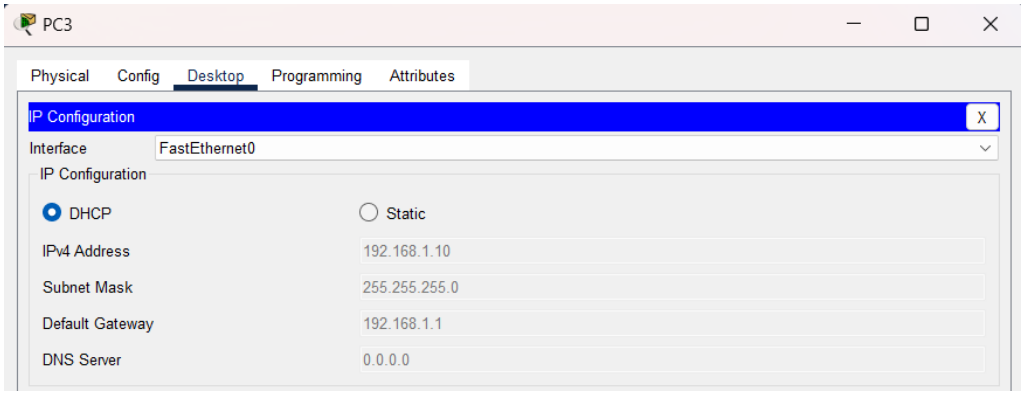


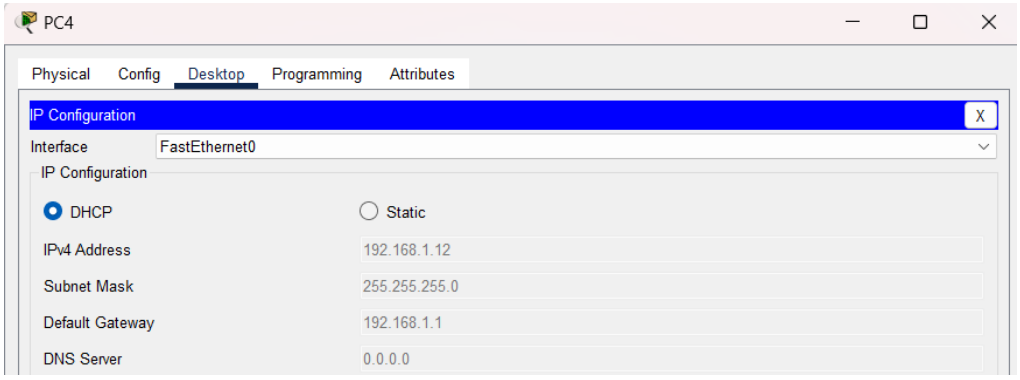
AUTOMATIC IP ALLOCATION IN PC's



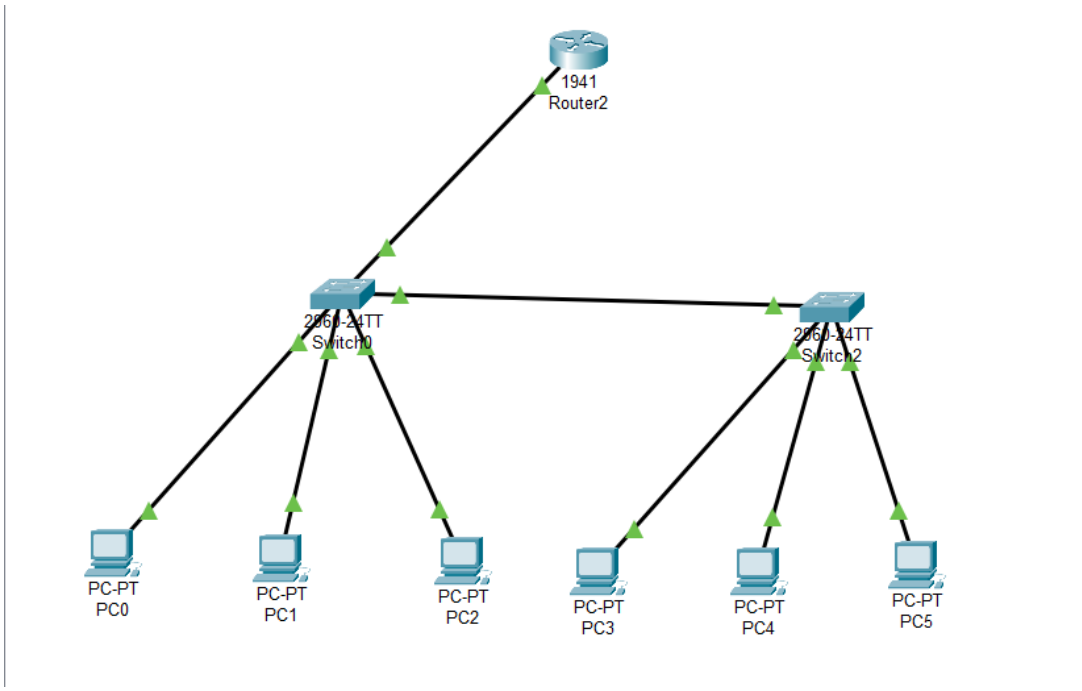



PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.11
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0

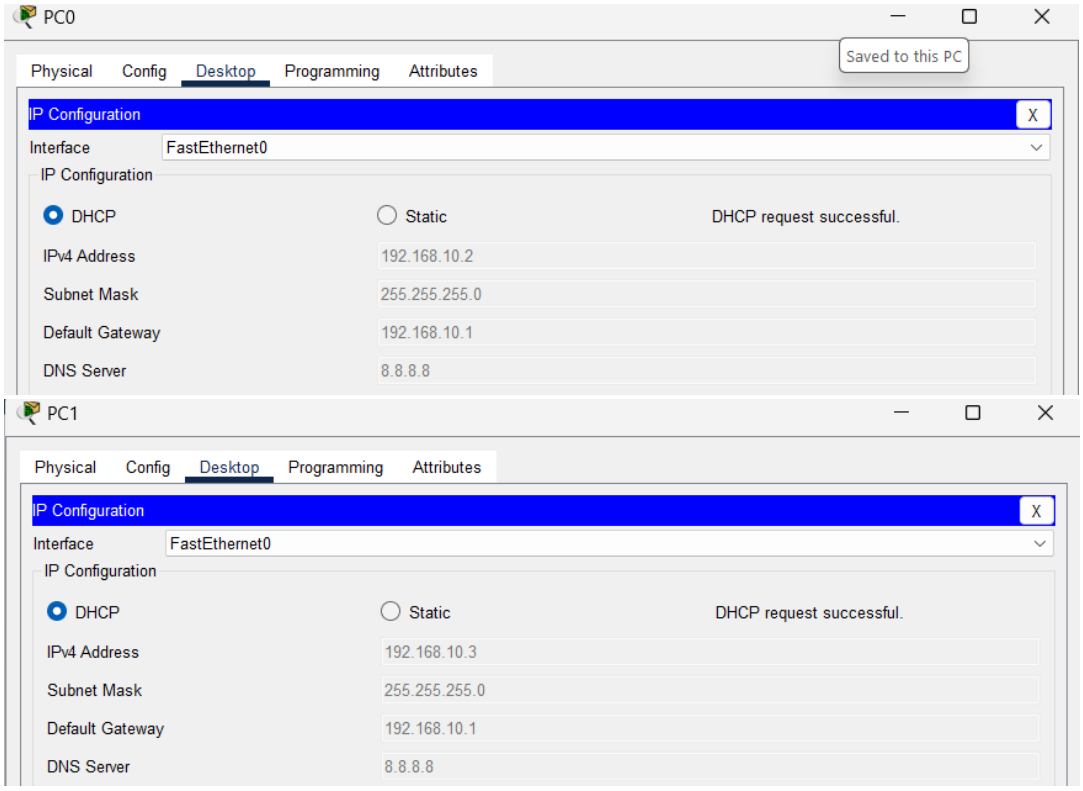

PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0

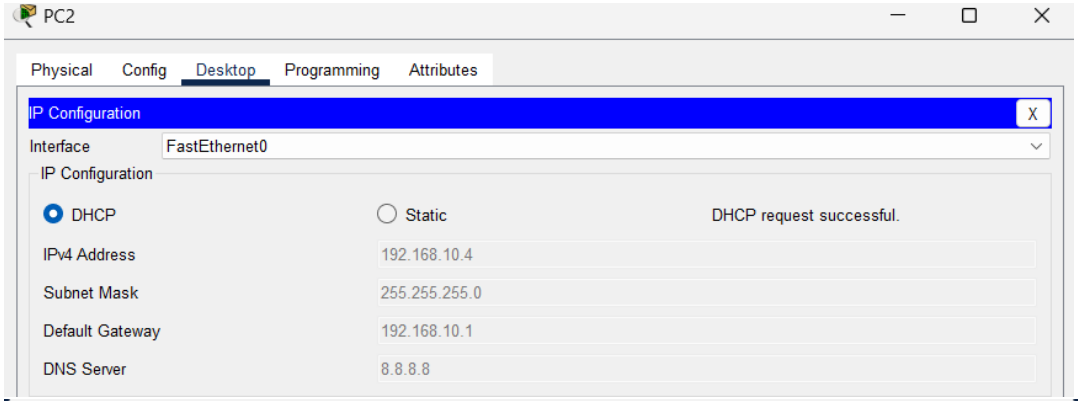

PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.12
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0

ii. MULTIPLE SUBNET NETWORKING:



AUTOMATIC IP ALLOCATION IN PC's




PC2
Physical
Config
Desktop
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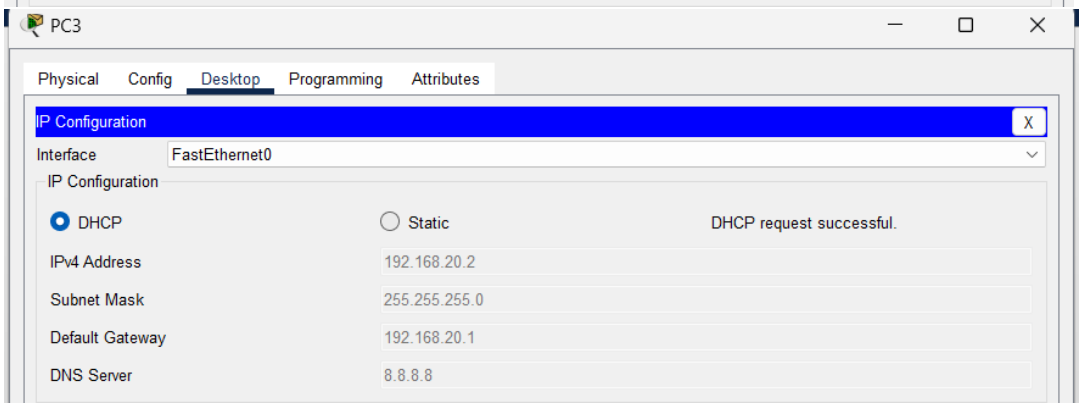
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.10.4
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
8.8.8.8


PC3
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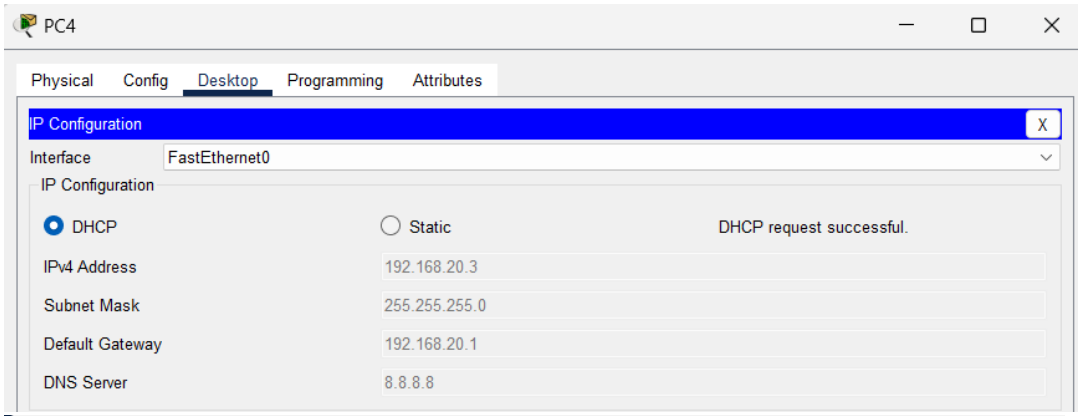
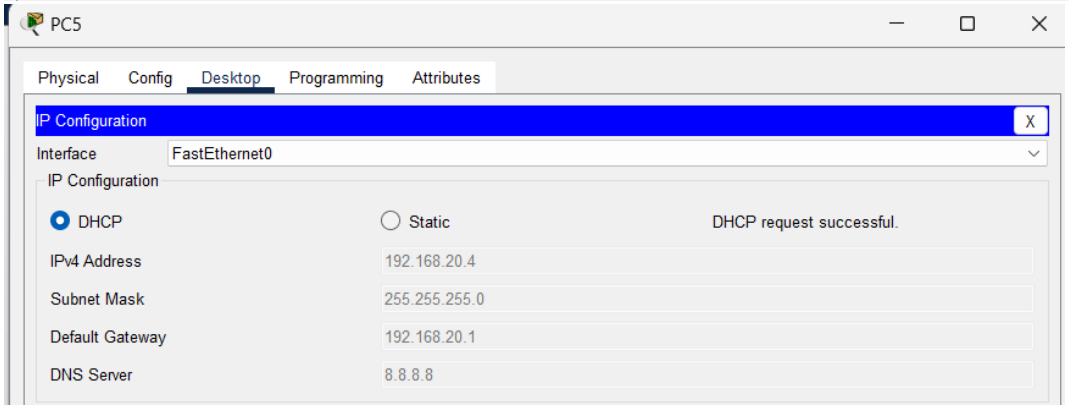
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.20.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
8.8.8.8


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.20.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
8.8.8.8


PC5
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.20.4
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
8.8.8.8

4.7 OUTPUT TESTING

i. SINGLE SUBNET NETWORKING:

GATEWAY PINGING

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::207:ECFF:FE10:8CC0
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.1.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=5ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms
```

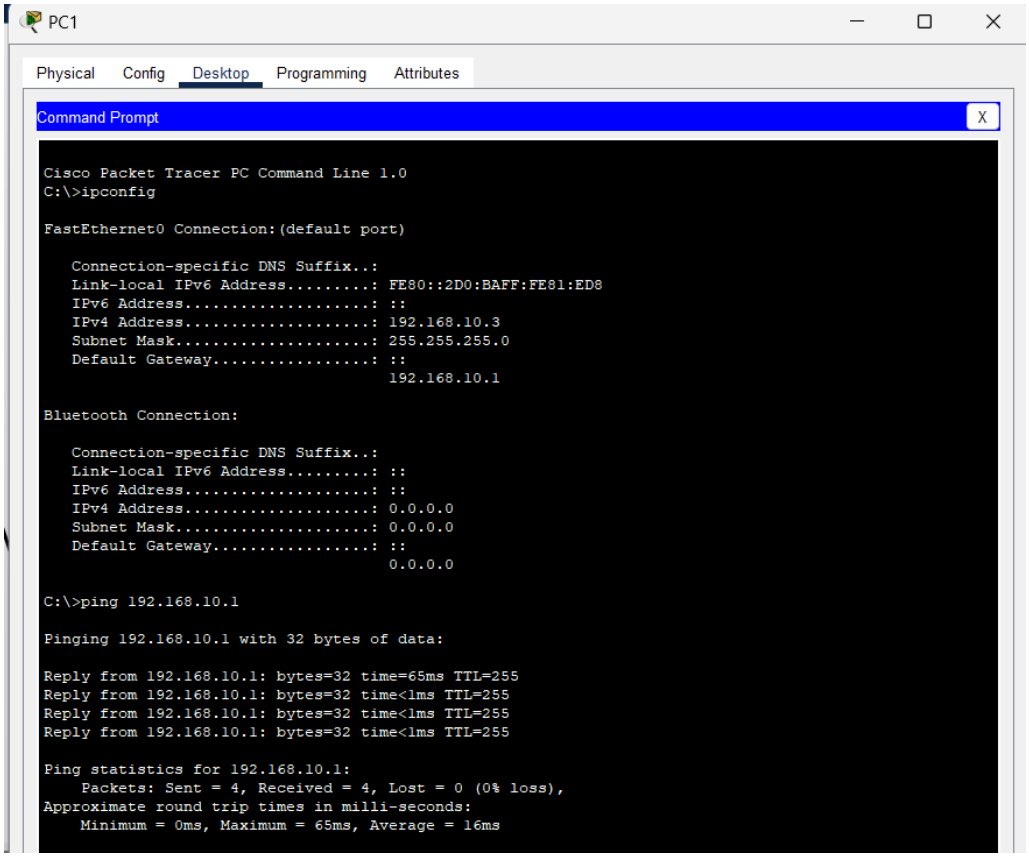
ii.MULTIPLE SYBNET NETWORKING

VLAN's ASSIGNMENT VERIFICATION

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   SALES                            active    Fa0/3, Fa0/4, Fa0/5
20   MARKETING                        active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

GATEWAY PINGING



5. SYSTEM IMPLEMENTATION

Implementation is the process of having systems personnel check out and put new equipment into use, train users, install the new application depending on the size of the organization that will be involved in using the application and the rsk associated with its use, system developers may choose to test the operation in only one area of the firm, say in one department or with only one or two persons. Sometimes they will run the old and new systems together to compare the results. In still other situation, developers will stop using the old system one-day and begin using the new one the next. As we will see, each implementation strategy has its merits, depending on the business situation in which it is considered. Regardless of the implementation strategy used,developers strinve to ensure that the system's initial use is trouble-free.

Objective

To configure a DHCP server using a Cisco router in Packet Tracer to automatically allocate IP addresses to devices in single-subnet and multi-subnet VLAN scenarios.

---

1. Single Subnet Topology Setup

Components Needed:

- 1 Cisco Router (1941 ISR)
- 1 Cisco Switch (2960)
- 5 PCs

Steps:

Step 1: Create the Network Layout

1. **Drag and drop components**:
    - Add 1 router, 1 switch, and 5 PCs to the workspace.
2. **Connect devices**:
    - Use copper straight-through cables:

- Connect the Router's **GigabitEthernet 0/0** to the Switch's **FastEthernet 0/1**.
- Connect PCs to the Switch's ports **f0/2 to f0/6**.

Step 2: Configure the Router as a DHCP Server

1. Click the Router and open the **CLI tab**.
2. Enter the following commands:

```
Router> enable
Router# configure terminal
Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.9
Router(config)# ip dhcp pool LAN_POOL
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.1.1
Router(dhcp-config)# dns-server 8.8.8.8 8.8.4.4
```

Step 3: Assign an IP Address to the Router Interface

1. Configure the router interface:

```
Router(config)# interface g0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
```

Step 4: Configure PCs to Obtain IP Addresses Dynamically

1. For each PC:
    - Open **Desktop > IP Configuration**.

- Select **DHCP** to obtain an IP address automatically.

Step 5: Verify the Configuration

1. On each PC, open the **Command Prompt** and type:

Copy code

```
ipconfig
```

- Confirm the assigned IP address and default gateway.

2. Ping the router's default gateway:

Copy code

```
ping 192.168.1.1
```

---

2. Multi-Subnet VLAN Topology Setup

Components Needed:

- 1 Cisco Router (1941 ISR)
- 2 Cisco Switches (2960)
- 6 PCs (3 in VLAN 10, 3 in VLAN 20)

Steps:

Step 1: Create the Network Layout

1. **Drag and drop components**:
   - Add 1 router, 2 switches, and 6 PCs to the workspace.
2. **Connect devices**:
   - Connect **Router g0/0** to **Switch 1 f0/1**.

- Create a **trunk link** between **Switch 1 f0/2** and **Switch 2 f0/2**.
- Connect PCs to their respective VLANs:
  - VLAN 10 PCs to **Switch 1**.
  - VLAN 20 PCs to **Switch 2**.

Step 2: Configure VLANs on the Switches

Switch 1:

```
Switch1> enable
Switch1# configure terminal
Switch1(config)# vlan 10
Switch1(config-vlan)# name SALES
Switch1(config)# vlan 20
Switch1(config-vlan)# name MARKETING
Switch1(config)# interface range f0/3 - 5
Switch1(config-if-range)# switchport mode access
Switch1(config-if-range)# switchport access vlan 10
Switch1(config)# interface f0/2
Switch1(config-if)# switchport mode trunk
```

Switch 2:

```
Switch2> enable
Switch2# configure terminal
Switch2(config)# vlan 10
Switch2(config-vlan)# name SALES
Switch2(config)# vlan 20
```

```
Switch2(config-vlan)# name MARKETING

Switch2(config)# interface range f0/3 - 5

Switch2(config-if-range)# switchport mode access

Switch2(config-if-range)# switchport access vlan 20

Switch2(config)# interface f0/2

Switch2(config-if)# switchport mode trunk
```

Step 3: Configure the Router as a DHCP Server

1. Configure DHCP pools:

```
Router> enable

Router# configure terminal

Router(config)# ip dhcp pool VLAN_10_POOL

Router(dhcp-config)# network 192.168.10.0 255.255.255.0

Router(dhcp-config)# default-router 192.168.10.1

Router(dhcp-config)# dns-server 8.8.8.8

Router(config)# ip dhcp pool VLAN_20_POOL

Router(dhcp-config)# network 192.168.20.0 255.255.255.0

Router(dhcp-config)# default-router 192.168.20.1

Router(dhcp-config)# dns-server 8.8.8.8
```

Step 4: Configure Sub-Interfaces on the Router

1. Configure sub-interfaces for VLANs:

```
Router(config)# interface g0/0.10

Router(config-subif)# encapsulation dot1Q 10
```

```
Router(config-subif)# ip address 192.168.10.1 255.255.255.0

Router(config-subif)#exit

Router(config)# interface g0/0.20

Router(config-subif)# encapsulation dot1Q 20

Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

Step 5: Configure PCs to Obtain IP Addresses Dynamically

1. For each PC:
    - Open **Desktop > IP Configuration**.
    - Select **DHCP**.

Step 6: Verify the Configuration

1. Run **ipconfig** on each PC to verify the assigned IP address.
2. Ping the default gateways:

```
ping 192.168.10.1

ping 192.168.20.1
```

3. Test inter-VLAN routing if configured.

---

6. MIKROTIK ROUTER BASIC CONFIGURATION

Instruction

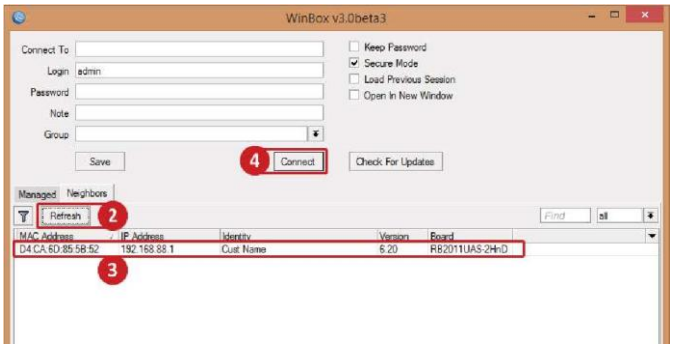
To access the Mikrotik, please change your network settings to be able to access 192.168.88.1 which is the default IP address of the Mikrotik.

Once the network settings have been updated, please connect to **ETH2** or **port 2** to access the **LAN** connection. Please note that **ETH1** or port 1is for the **WAN** connection.

With the network settings able to access 192.168.88.1 and your computer is connected to eth2, please open a web browser and enter in 192.168.88.1 in the address bar.

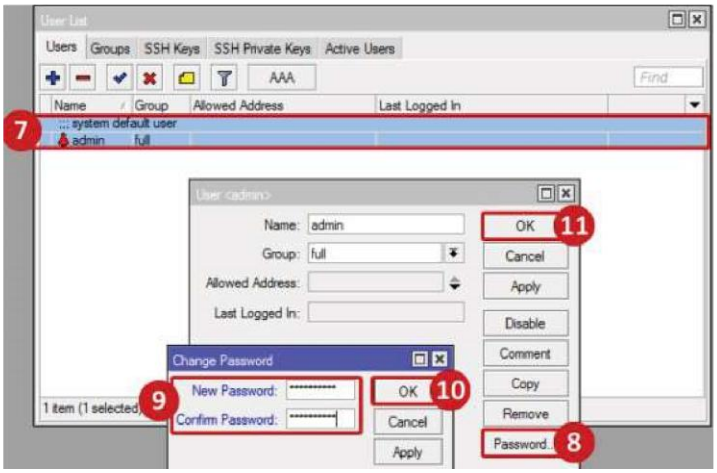
You will now have the web login screen available. Here will want to download Winbox**,** which is the tool we will be using to manage the Mikrotik.

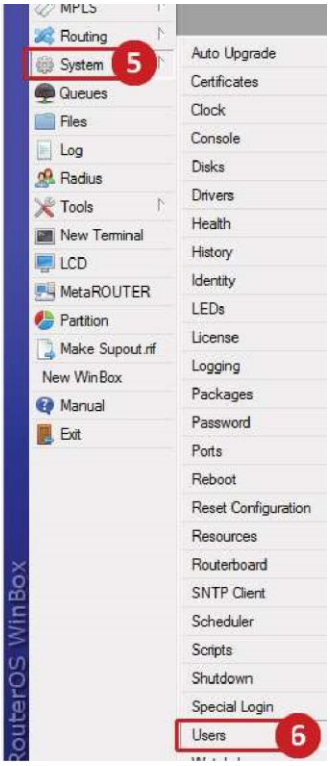
1. From the Mikrotik login web-page, click on **Winbox** to download the Winbox application.
2. Once Winbox has been downloaded, open the application and it will find the Mikrotik, if not press **Refresh** to find the Mikrotik.
3. The Mikrotik will display the Mikrotik with the **MAC Address, IP Address, Version,** and the **Board.**
4. By default the login is admin and there is no password. Press **Connect** to connect the Mikrotik.



Set Password for Admin Account

5. Go to System
6. Go to Users
7. From the User List, select the admin account by double clicking the account
8. From the User <admin> window, click on Password
9. From the Change Password window, enter in the desired password and re-enter to confirm
10. Click OK from the Change Password window
11. Click OK from the User <admin> window





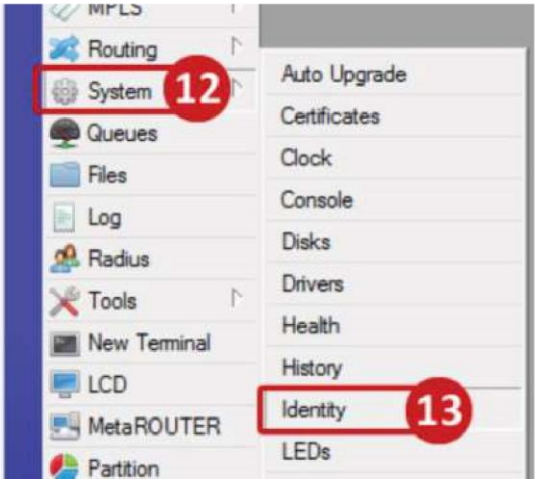
Set Identity

12.Go to System

13.Go to Identity

14.In the Identity window, enter in an identity to identify the given Mikrotik (i.e.: Cust Name)

15.Click OK



WinBox installation

WinBox is a small utility that allows the administration of MikroTik RouterOS using a fast and simple GUI.

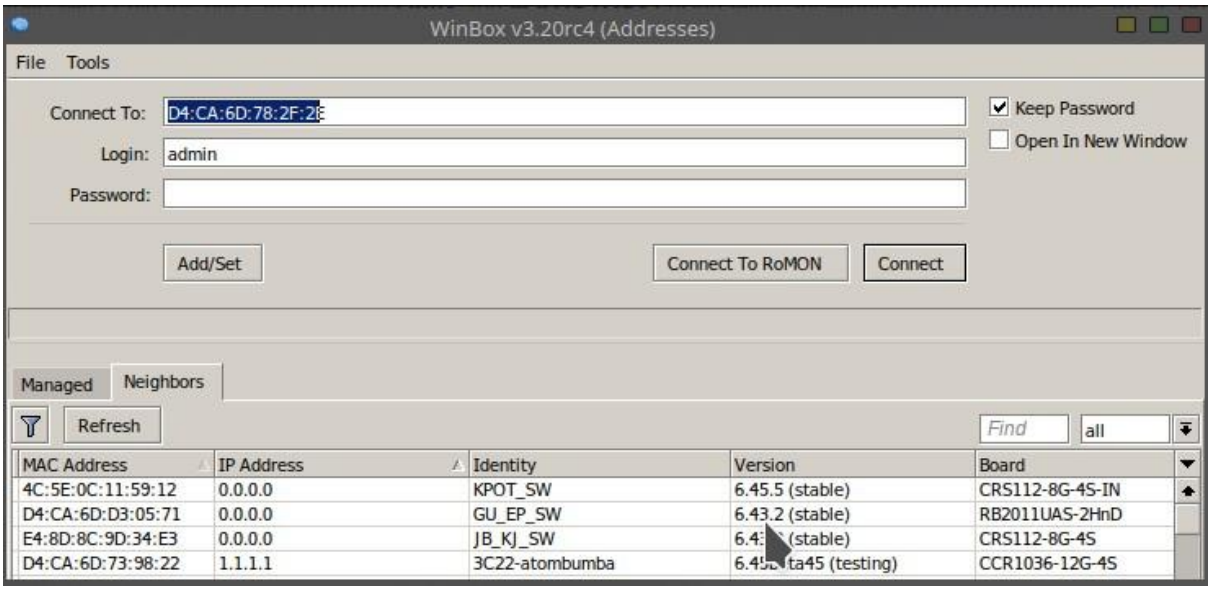
Mikrotik Hap Lite

WinBox loader can be downloaded from the MikroTik download page. https://mikrotik.com/download

When WinBox.exe is downloaded, double click on it, and the WinBox loader window will pop up. There are two WinBox loader modes: simple which is enabled by default and advanced.

Simple mode

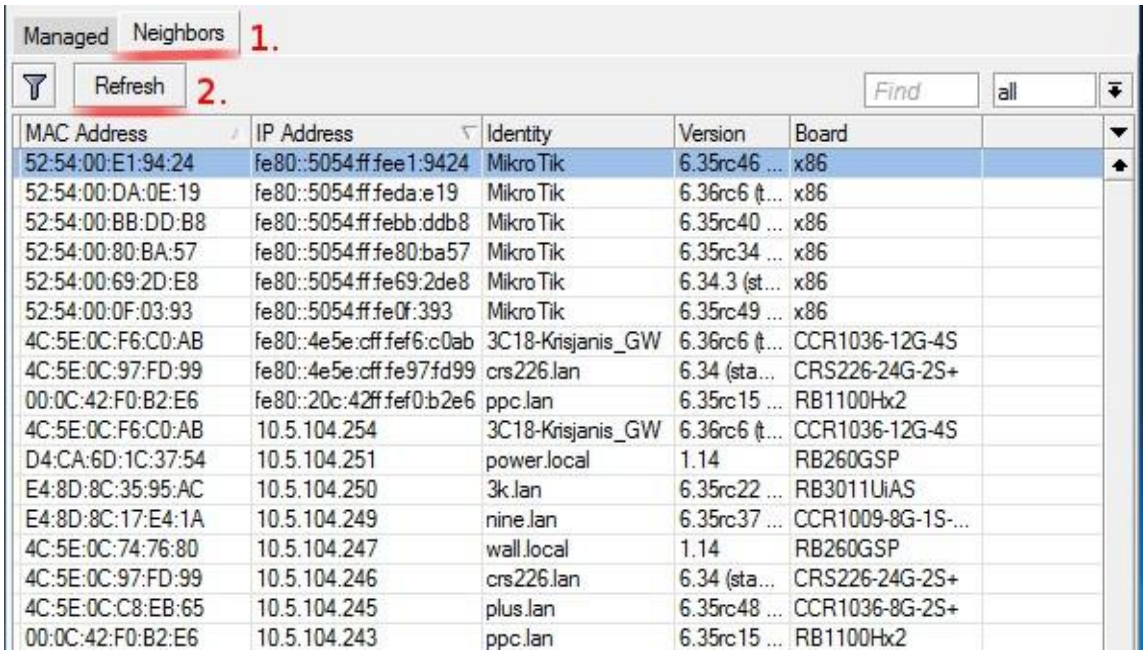
When you open WinBox loader for the first time simple mode layout will be used:



To connect to the router enter the IP or MAC address of the router, specify username and password (if any) and click on the **Connect** button. You can also enter the port number after the IP address, separating them with a colon, like this 192.168.88.1:9999. The port can be changed in the RouterOS **services** menu.

It is recommended to use an IP address whenever possible. MAC session uses network broadcasts and is not 100% reliable.

You can also use neighbor discovery, to list available routers use the **Neighbors** tab:



From the list of discovered routers, you can click on the IP or MAC address column to connect to that router. If you click on IP address then IP will be used to connect, but if you click on MAC Address then the MAC address will be used to connect to the router.

Neighbor discovery will show also devices that are not compatible with WinBox, like Cisco routers or any other device that uses CDP (Cisco Discovery Protocol). If you will try to connect to a SwOS device, then the connection will be established through a web browser

Buttons/check-boxes and Other Fields

- **Connect** - Connect to the router
- **Connect To RoMON** - Connect to RoMON Agent
- **Add/set** - Save/Edit any of the saved router entries in the **Managed** tab.
- **Open In New Window** - Leaves loader open in the background and opens new windows for each device to which connection is made.
- **Connect To:** - destination IP or MAC address of the router
- **Login** - username used for authentication
- **Password** - password used for authentication
- **Keep Password** - if unchecked, the password is not saved to the list

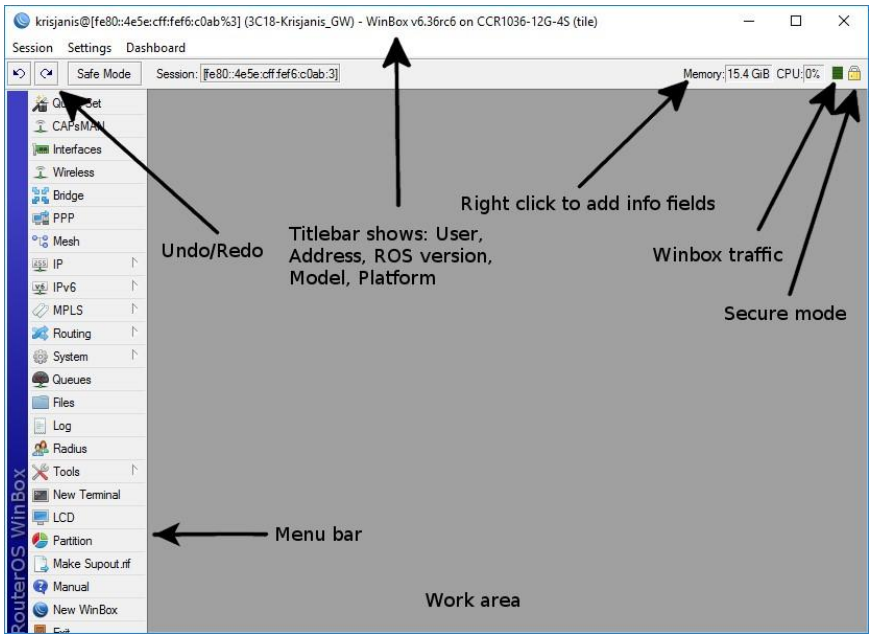
Menu Items

- **File**
- **New** – Create a new managed router list in a specified location
- **Open –** Open managed router list file
- **Save As –** Save current managed router list to file
- **Exit** – Exit WinBox loader
- **Tools**
- **Advanced Mode** – Enables/Disables advanced mode view
- **Import** - Imports saved session file
- **Export** - Exports saved session file
- **Move Session Folder** - Change path where session files are stored
- **Clear cache** - Clear WinBox cache
- **Check For Updates** - Check for updates for WinBox loader

Interface Overview

WinBox interface has been designed to be intuitive for most of the users. The interface consists of:

- The main toolbar at the top where users can add various info fields, like CPU and memory usage.
- The menu bar on the left - list of all available menus and sub-menus. This list changes depending on what packages are installed. For example, if the IPv6 package is disabled, then the **IPv6** menu and all its sub-menus will not be displayed.
- Work area - an area where all menu windows are opened.



The title bar shows information to identify with which router WinBox session is opened. Information is displayed in the following format:

[username]@[Router's IP or MAC] ([RouterID]) - WinBox [ROS version] on [RB model] ([platform])

From screenshot above we can see that user **krisjanis** is logged into router with IPv4/IPv6 address **[fe80::4e5e:cff:fef6:c0ab%3]**. Router's ID is **3C18-Krisjanis_GW**, currently installed RouterOS version is **v6.36rc6**, RouterBoard is **CCR1036-12G-4S** and platform is **tile**.

On the Main toolbar's left side is located:

- **undo**
- **redo**
- **Safe Mode**
- Currently loaded session

7. CONCLUSION

My project completion was an enriching experience that significantly enhanced my understanding of network design and deployment. The skills and knowledge I gained have prepared me for future roles in creating and implementing robust communication systems. Network design and deployment are fundamental to building secure, efficient, and scalable communication infrastructures for both organizations and individuals. Effective design ensures a network can support operational needs while accommodating growth and evolving technologies. This process starts with identifying the network's requirements, followed by strategic planning of its topology, hardware, software, and security measures. The architecture—be it client-server, peer-to-peer, or hybrid—is chosen based on the network's purpose and complexity. Key considerations such as bandwidth, latency, redundancy, and fault tolerance are essential for achieving optimal performance. The deployment phase involves setting up hardware like routers, switches, and servers, configuring software and protocols, and conducting rigorous testing and troubleshooting to ensure seamless operation. Security is a top priority throughout, with measures such as encryption, firewalls, and intrusion detection systems safeguarding the network against potential threats. Best practices, including access controls, VPNs, and regular audits, help mitigate risks and protect sensitive data. Post-deployment maintenance and monitoring are equally critical to ensure long-term performance and adaptability. This includes periodic upgrades, capacity planning, and proactive responses to emerging security challenges. The dynamic nature of technology makes ongoing evaluation and improvement indispensable in maintaining a functional and secure network infrastructure.